

Revenue Reconciliation & Leakage Analysis

1. Executive Summary

This report presents a structured analysis of adidas transactional sales data (kaggle) to identify revenue leakage driven by mismatched sales values, pricing inconsistency, and margin inefficiencies. Sales were recalculated using unit price and quantity to validate reported revenue. The analysis confirms measurable revenue loss and identifies operational drivers contributing to the issue.

2. Problem Statement

The company wants to ensure that

1. Reported revenue is accurate.
2. Pricing and Profitability are consistent.
3. There is no hidden revenue or margin leakage.

3. Business Objectives

1. Validate revenue accuracy at invoice level.
2. Identify pricing inconsistencies.
3. Evaluate sales method profitability.
4. Assess regional margin consistency.

4. Hypothesis (Possible Outcomes)

1. Total Sales may not match Price * Units.
2. Inconsistent pricing across regions or sales methods.
3. Some sales methods have lower operating margins despite similar revenue.
4. Certain regions show weaker margins.

5. Dataset Description

The dataset includes the following fields: (Adidas Sales from kaggle website)

- Retailer, Retailer ID
- Invoice Date
- Region, State, City
- Price per Unit
- Units Sold
- Total Sales (Reported)
- Operating Profit
- Sales Method

6. Methodology

1. Converted numeric columns from string to numeric format.
2. Removed currency symbols and formatting inconsistencies.
3. Recalculated sales using Price per Unit × Units Sold.
4. Computed revenue variance between reported and recalculated sales.
5. Flagged mismatched transactions.
6. Calculated price deviation percentage.
7. Computed operating margin across regions and methods.
8. Developed BI dashboards for executive reporting and root cause analysis.

7. Tools Used

1. Python – Data cleaning and validation.
2. SQL – Aggregation and hypothesis testing.
3. Power BI – Dashboard visualization.
4. ChatGPT – Concept clarification and productivity enhancement.

8. SQL& Python Analysis Evidence

SQL queries were used to aggregate revenue variance, calculate margin differences, and analyze price deviation across dimensions.

Screenshot Placeholder – SQL Queries

```
5      # HYPOTHESIS RELATED TO REVENUE LEAKAGE
6      #1. Is there a revenue mismatch in the data?
7      • Select `Revenue Flag`,
8          count(*) as invoice_count,
9          sum(`Calculated Sales`) as "Total Sales" from adidas_sales
10     group by `Revenue Flag`;
11
```

Revenue Flag	invoice_count	Total Sales
Mismatch	5760	33590125
Correct	3881	86395900

```
# RELATED TO PRICING AND MARGIN EFFICIENCY
#7. Some products are sold at inconsistent prices across regions or sales methods
Select Product, Region, `Sales Method`, round(avg(`Price per Unit`), 2) As Avg_price,
round(avg(`Price Per Unit`) - avg(avg(`Price Per Unit`)) over (Partition by Product), 2) as Price_Deviation
from adidas_sales
group by Product, Region, `Sales Method` ;

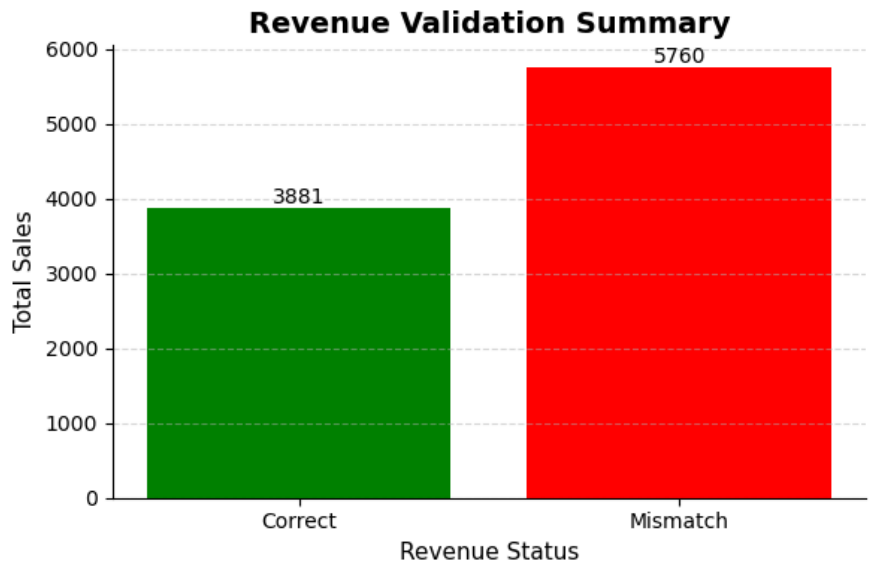
#8. Some sales methods have lower operating margins despite similar revenue
select `Sales Method`,sum(`Operating Profit`) As Total_Profit, sum(`Calculated Sales`) As Total_Sales,
round(sum(`Operating Profit`) * 100.0 / Nullif(sum(`Calculated Sales`), 0), 2) As Operating_Margin
from adidas_sales
group by `Sales Method`;

#9. Certain regions show weaker margins for the same product
select Product, Region,
Round(sum(`Operating Profit`) * 100.0 / Nullif(sum(`Calculated Sales`), 0), 2) As Operating_Margin
from adidas_sales
group by Product, Region;
```

Python Analysis & Evidence

Python is used for data cleaning and data validation and small graphical representation of mismatch vs recalculated (Correct) sales.

Screenshot Placeholder



9. Key Results

- Total Reported Sales: 89.82M
- Total Corrected Sales: 119.99M
- Revenue Loss: -30.17M
- Revenue Loss Percentage: 33.58%
- Average Price per Unit: 45.21
- Price Deviation: 119.82%
- Operating Margin: 27.63%

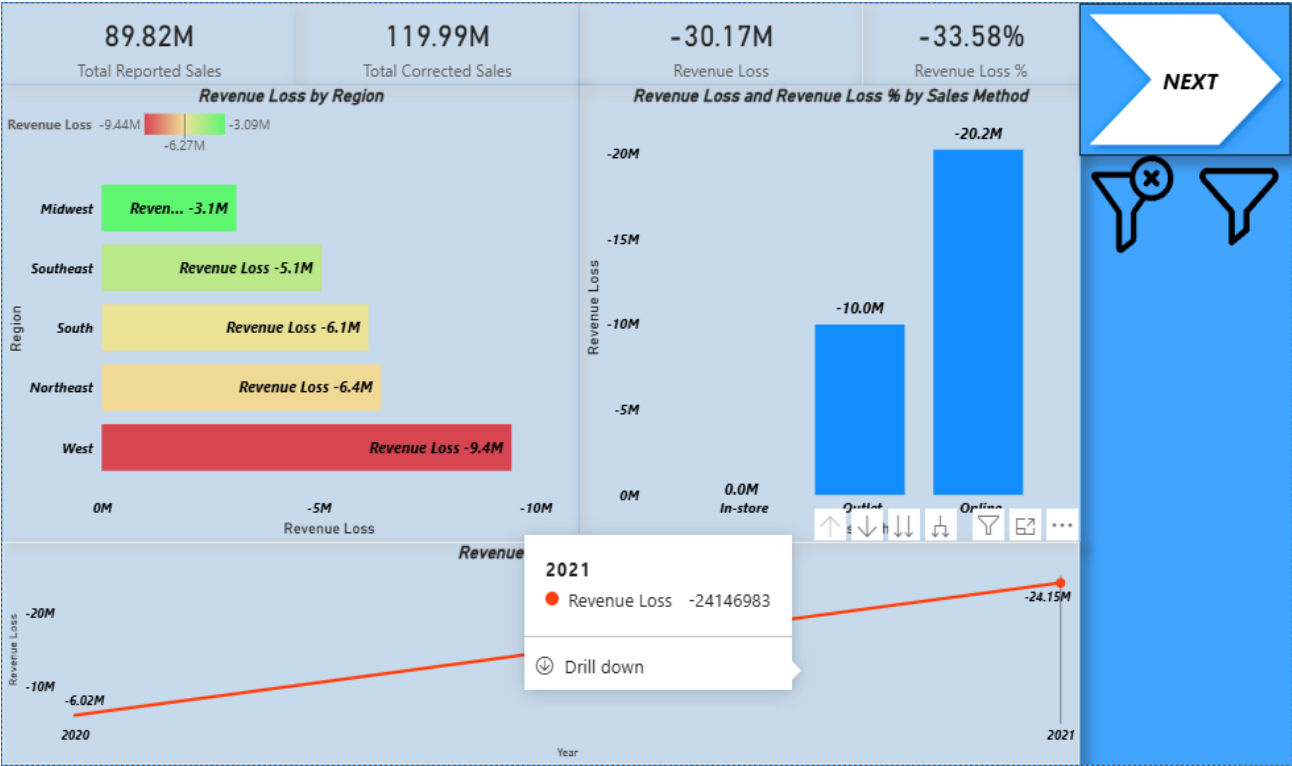
The 119.82% price deviation exists despite clean data and no outliers, confirming pricing inconsistency rather than data quality errors. Revenue leakage of 33.58% represents a material financial impact.

10. Dashboard Structure and Justification

6.1 Dashboard Page 1 – Executive Summary

This page quantifies the magnitude of revenue leakage using KPIs such as total reported sales, corrected sales, revenue loss, leakage percentage, and operating margin. It establishes the financial seriousness of the issue.

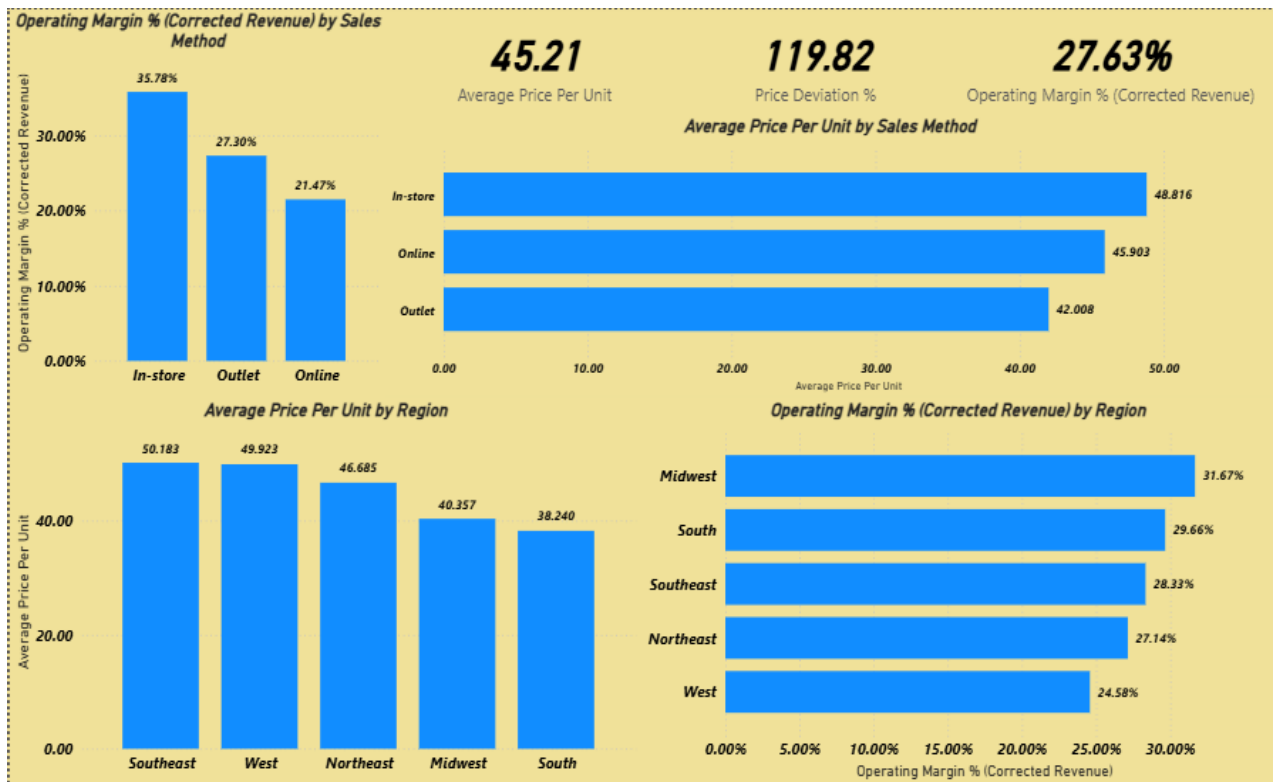
Screenshot Placeholder – Dashboard Page 1



6.2 Dashboard Page 2 – Pricing & Margin Drivers

This page identifies the operational causes of leakage by analyzing price deviation and operating margin differences across regions and sales methods. It confirms the hypotheses related to pricing inconsistency and margin inefficiency.

Screenshot Placeholder – Dashboard Page 2



11. Hypothesis Validation

- Hypothesis 1: Products are sold at inconsistent prices across regions or methods.
Result: 119.82% deviation confirms inconsistency.
- Hypothesis 2: Some sales methods have lower operating margins despite similar revenue.
Result: Margin comparison shows variation.
- Hypothesis 3: Certain regions show weaker margins.
Result: Regional margin variation confirms inefficiency.

12. Conclusion

Revenue leakage is confirmed and measurable. The issue is not caused by missing or invalid data but by pricing inconsistency and operational inefficiencies. Stronger pricing governance and margin monitoring controls are required to reduce financial loss.